

Generation Interconnection System Impact Study Report

For Queue Project Q16-0563

Longdon 138 kV
January 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0563. The Interconnection Customer proposes a 159.6 MW wind facility interconnecting at the Longdon 138 kV substation in the Kentucky area. The request was received on January 01, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 195 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,979,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Leslie Substation-St Joseph 138 KV Line	Albion Road-Wixom 138 KV	174.6	221	28.9
Leslie-N Appleton 138 KV	Wixom-Albion Road 138 KV	174.6	224	28.9
Longdon-Heybourne 138 KV Line	Barnabie Cable Sta-Longdon Substation 138 KV	152.8	231	100.0
Wellington-Davisville 138 KV line	Wixom-Albion Road 138 KV	147.3	197	19.1
Ayers Island-Wixom 138 KV	Davisville-N Appleton 138 KV Line	126.7	257	32.5
St Joseph-Chestnut 138 KV	Wixom-Albion Rd 138 KV line	126.5	295	28.9
Mirabel-Wec 138 KV	Wellington-Wec 138 KV	123.9	222	21.7
Davisville-Mirabel 138 KV line	Wec-Wellington 138 KV line	123.6	217	21.7
Davisville-North Appleton 138 KV	Wixom Substation-Albion Rd 138 KV	121.6	329	28.9
Campbell Jct-School St Substation 138 KV	Glendale S Tap 6-Campbell Jct 138 KV L	121.3	212	30.7
Ubc South Campus Substation-Danz Ave Sw Sta Substation 138 KV Line	Albion Road-Platte City Substation 138 KV	121.0	192	12.9
51St St-Anson N 138 KV L	Wixom Substation-Albion Rd 138 KV Line	119.1	227	20.2
51St St-Culver 138 KV	Albion Rd-Wixom 138 KV	117.7	214	20.2
Longdon-Barnabie Cable Sta 138 KV Line	Moss Bluff-Barnabie Cable Sta 138 KV Line	115.1	366	68.9
Sand Springs-Ojo 138 KV	Sand Springs-Snohmish 138 KV	113.9	232	13.4
Cwec Tigerton-Danz Ave Sw Sta 138 KV L	Albion Rd-Platte City 138 KV	111.3	202	12.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wellington-Kaweah 1 138 KV	Davisville-N Appleton 138 KV	109.3	237	22.0
Mossyrock Substation-Washington County Cogeneration Facility 230 KV	Wolfe-Barrett 500 KV	109.2	739	5.2
N Hendrickson-Clubhouse 138 KV L	Glendale S Tap 6-Campbell Jct Substation 138 KV line	109.0	216	30.7
Ayers Island-Kaweah 1 138 KV	N Appleton-Davisville 138 KV L	108.0	236	22.0
Lovett-Barnabie Cable Sta 138 KV	Barnabie Cable Sta-Casino 138 KV	107.2	292	38.1
Fernwood-Chestnut 138 KV	Consolidated Coal-Chestnut 138 KV	107.1	202	21.1
Wec-Wellington 138 KV	Barnabie Cable Sta-Casino 138 KV line	106.0	370	34.0
Cwec Tigerton-Ellendale Substation 138 KV L	Platte City-Albion Road 138 KV line	104.4	209	12.9
Cedar Tap-Idaho Falls Power 138 KV	Barnabie Cable Sta-Casino 138 KV Line	102.7	355	38.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Leslie Substation-St Joseph 138 KV Line	Rebuild line with higher-capacity conductor	\$9,080,000	11% (\$1,009,000)
Lslie-North Appleton 138 KV	Rebuild line with higher-capacity conductor	\$8,323,000	11% (\$925,000)
Longdon-Heybourne 138 KV Line	Rebuild line with higher-capacity conductor	\$3,370,000	42% (\$1,422,000)
Wellington-Davisille 138 KV line	Rebuild line with higher-capacity conductor	\$4,490,000	11% (\$486,000)
Ayers Island-Wixom 138 KV	Rebuild line with higher-capacity conductor	\$16,502,000	14% (\$2,232,000)
St Joseph-Chestnut 138 KV	Rebuild line with higher-capacity conductor	\$3,957,000	11% (\$440,000)
Wec-Mirabel 138 KV	Rebuild line with higher-capacity conductor	\$4,562,000	18% (\$835,000)
Davisville-Mirabel 138 KV line	Rebuild line with higher-capacity conductor	\$24,286,000	18% (\$4,443,000)
Davisville-North Appleton 138 KV	Rebuild line with higher-capacity conductor	\$6,979,000	11% (\$775,000)
School St-Campbell Junction 138 KV	Rebuild line with higher-capacity conductor	\$4,061,000	17% (\$705,000)
Ubc South Campus Substation-Danz Ave Sw Sta Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$9,884,000	10% (\$959,000)
51St Street-Anson North 138 KV	Rebuild line with higher-capacity conductor	\$24,438,000	14% (\$3,518,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
51St St-Culver 138 KV	Rebuild line with higher-capacity conductor	\$5,461,000	14% (\$786,000)
Longdon-Barnabie Cable Sta 138 KV Line	Rebuild line with higher-capacity conductor	\$11,409,000	31% (\$3,575,000)
Sand Springs-Ojo 138 KV	Rebuild line with higher-capacity conductor	\$13,945,000	10% (\$1,433,000)
Cwec Tigerton-Danz Ave Sw Sta 138 KV	Rebuild line with higher-capacity conductor	\$5,475,000	10% (\$531,000)
Wellington-Kaweah 1 138 KV	Reconductor line	\$3,768,000	17% (\$624,000)
Mossyrock Substation-Washington County Cogeneration Facility 230 KV	Reconductor line	\$27,687,000	11% (\$2,956,000)
N Hendrickson-Clubhouse 138 KV L	Reconductor line	\$1,036,000	17% (\$180,000)
Ayers Island-Kaweah 1 138 KV	Reconductor line	\$10,855,000	17% (\$1,798,000)
Lovett-Barnabie Cable Sta 138 KV	Reconductor line	\$6,378,000	21% (\$1,354,000)
Fernwood-Chestnut 138 KV	Reconductor line	\$2,969,000	11% (\$333,000)
Wec-Wellington 138 KV	Reconductor line	\$15,339,000	19% (\$2,876,000)
Cwec Tigerton-Ellendale Substation 138 KV L	Reconductor line	\$1,672,000	10% (\$162,000)
Cedar Tap-Idaho Falls Power 138 KV	Reconductor line	\$3,160,000	20% (\$622,000)
	Total allocated to project		\$34,979,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.